

Alireza Barzin Zanganeh

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SUMMARY

Senior AI Developer with over 10 years of software engineering experience across backend and identity platforms, specializing in AI-driven risk scoring and authentication systems. Skilled in designing and deploying generative AI and LLM solutions-including RAG and agentic pipelines-via FastAPI in Kubernetes-enabled cloud environments. Proven track record of enhancing model accuracy and reducing deployment time to deliver secure, scalable AI solutions for enterprise workflows.

SKILLS

- **Programming Languages & Frameworks:** Python, FastAPI, APIs
- **AI & Machine Learning:** Generative AI, AI solutions, large language models (LLMs), retrieval-augmented generation (RAG), NVIDIA technologies, NVIDIA NIM, NVIDIA NeMo, GPU-accelerated Training
- **DevOps & Infrastructure:** CI/CD pipelines, MLOps, Kubernetes, Docker, Containerized Environments
- **Cloud & Microservices Architecture:** cloud-based AI infrastructure, Cloud-native Architectures, Microservices
- **Engineering Domains:** Backend Engineering, Data Engineering, Identity & Security

EXPERIENCE

Consulting

2025 - Present

AI Platform Engineer

- Engineered Java-based Android SDK modules and enhanced cross-device digital fingerprinting workflows using NVIDIA technologies.
- Developed over 5 end-to-end ML and LLM applications leveraging generative AI, increasing model accuracy and deployment readiness by 30%.
- Architected agentic AI systems with tool-calling, multi-step reasoning, and retrieval-augmented generation (RAG) on GPU-accelerated training infrastructure, decreasing hallucinations by 50%.
- Designed ML-integrated authentication workflows and risk-scoring components, boosting fraud detection precision by 30%.
- Implemented CI/CD pipelines and containerized environments, reducing deployment time by 50% and increasing release reliability.

SecureAuth

2022 - 2025

Senior Software Engineer - ML Systems & Backend

- Engineered ML-integrated authentication workflows and risk-scoring components, improving fraud detection precision by 30%.
- Built microservices and APIs for identity, device trust, and anomaly detection across cloud-native architectures, reducing API latency by 30% through containerized environments.
- Developed evaluation and monitoring tools for ML-driven authentication systems, reducing model drift incidents by 35%.
- Implemented CI/CD pipelines (GitHub Actions, Jenkins) improving deployment reliability by 50%.
- Created Python-based validation and automation systems supporting ML-integrated authentication pipelines.

Acceptto

2016 - 2022

Software Engineer - Backend, Identity, ML Infrastructure

- Built backend services and SDK components for identity and behavioral authentication systems across mobile and web platforms.
- Developed Java-based Android SDK modules and enhanced cross-device digital fingerprinting workflows using NVIDIA technologies.

PROJECTS

Smart Resume Agent - AI Resume Tailoring Platform

- Built a four-phase LLM pipeline (keyword extraction → gap analysis → rewrite → QA) improving ATS alignment by 15–30 points.
- Implemented LLM provider abstraction (Gemini, Claude, OpenAI, BYOK) reducing integration overhead by 70%.
- Designed embedding-grounded rewriting using pgvector, eliminating hallucinated experience.
- Created inline editing, version history, and patch-based chat edits, reducing rewrite cycles by 50%.
- Deployed platform as containerized microservices using FastAPI, Docker, and Kubernetes with CI/CD pipelines to ensure scalable, secure production release

Movie Agent Service - Agentic AI for Movie Discovery

- Designed a Python-based tool-calling agent architecture with strict separation of reasoning and execution
- Implemented RAG with FAISS, improving retrieval relevance by 40%.
- Built multi-tool orchestration (search, statistics, quiz generation, vision analysis).
- Added validation layers and structured error handling, reducing failure cases by 60%.
- Deployed microservices using FastAPI and Docker, orchestrated with Kubernetes and integrated CI/CD pipelines for continuous delivery

ASAR - Device Identity, Fingerprinting & Risk Engine (Private)

- Built device identity + fingerprinting pipeline improving device recognition accuracy by 35%.
- Implemented OIDC discovery, JWKS, and secure internal token issuance.

- Designed anomaly detection engine for behavioral and device-level signals.
- Added Argon2id password storage, account lockout, and hardened login flows.

Rai - Agentic AI Layer for Identity Workflows (Private)

- Built domain-specialist agents with RBAC-scoped tools and fail-closed safety gates.
- Implemented retrieval-augmented generation with structured action flows.
- Added 570+ golden eval cases with CI-gated releases, reducing regressions by >80%.

Churn Risk Intelligence

- Built churn prediction models with 15–20% improvement in recall for high-risk segments.
- Designed feature pipelines capturing usage, tenure, and contract behavior.

Bank Term Deposit Prediction

- Improved minority-class recall by 30–45% using SMOTE + threshold tuning.
- Delivered interpretable models for business decision support.

Fraud Shield AI

- EDA, timezone-aware feature engineering, scalable PySpark preprocessing, supervised and deep learning models, ensemble risk tiering, and a Streamlit inference app.
- Tech: Python, PySpark, scikit-learn, XGBoost, PyTorch, Optuna, Streamlit.
- GPU-accelerated training with PyTorch + CUDA for deep learning (MLP, ResNet, LSTM) and tabular transformers (FT-Transformer, self-attention MLP) on an NVIDIA RTX 5070 Ti.
- GPU-accelerated inference for batch scoring and interactive deployment via trained PyTorch checkpoints and the production Streamlit app.
- Hands-on with the NVIDIA GPU stack (CUDA drivers, conda/PyTorch CUDA builds, Docker) as the foundation for NVIDIA NIM (LLM inference serving) and NeMo (LLM fine-tuning / RAG) on local Ubuntu.
- Leak-free, time-aware feature engineering with point-in-time Spark window functions (~30 features).
- XGBoost + ensemble (soft voting / stacking) with a two-stage risk system (Auto-Block / Review / Cleared).
- Honest temporal evaluation across train/val/test splits to reflect real-world fraud drift.

EDUCATION

IK

Software Engineering Program, Machine Learning, Deep Learning, LLMs, MLOps, System Design

- **Achievements:** Capstone Project: Fraud Shield AI (end-to-end fraud detection pipeline) - 800/800
- **Coursework:** Machine Learning, Deep Learning, LLMs, MLOps, System Design

Industrial Management Institute

Executive MBA

Allameh Tabataba'i University

BSc, Economics

Portland Community College

Certificate, Web Development & Design

- **GPA:** 3.9